

Math Class 10 Federal Board Chapter

01 Complex Numbers Exercise 1.2

Developing skilled knowledge

- ✓ What is an imaginary number? $\rightarrow a+bi, b \neq 0$
- ✓ Is an imaginary number real number? $\rightarrow \text{NO}$
- ✓ Provide examples of numbers that are pure imaginary numbers. $2i, -3i, 5i$
- ✓ What is the pattern formed by the powers of i ? $i, -1, -i, 1$
- ✓ Can we say product of $(2-3i)(2-3i)$ is imaginary? $\rightarrow \text{Yes}$
- ✓ If conjugate of z is $\bar{z}$, then is $z\bar{z}$ a real number? $\rightarrow \text{Yes}$

$$i^1 = i, \quad i^2 = -1, \quad i^3 = i^2 \cdot i = -i, \quad i^4 = (i^2)^2 = (-1)^2 = 1$$

$$i^5 = i^4 \cdot i = i$$

$$(2-3i)(2-3i) = (2-3i)^2 = -5-12i$$

$$z\bar{z} = \text{Real No.} \quad \text{Always}$$

Properties of Complex No.

Commutative Law

$$Z + W = W + Z$$

$$Z \times W = W \times Z$$

$$0 = 0 + 0i$$

Additive Identity of

Q. 1. Additive Identity of Complex No.

Multiplicative Identity of

Complex No. $1 \Rightarrow 1 + 0i$, $(1, 0)$
 $\uparrow \uparrow$
 Re Im

$$\mathbb{Z} \cdot 1 = 1 \cdot \mathbb{Z} = \mathbb{Z}$$

Associative Law

$$(Z+W) + V = Z + (W+V)$$

$$(Z \cdot W) V = Z \cdot (WV)$$

Distributive Property

$$Z(\vec{W} + \vec{V}) = Z\vec{W} + Z\vec{V}$$

$$Z(\underline{w-v}) = Zw - Zv$$

* Additive Inverse $5 + (-5) = 0$

*

ADDITION

$\neq 0$
Add. Identity

$$Z = a + bi$$

Additive Inverse $\Rightarrow -Z = -(a + bi)$
 $= \underline{\underline{-a - bi}}$

$$Z = 2 - 3i$$

$$\rightarrow ? \quad \boxed{-Z = -2 + 3i}$$

$$Z + (-Z) = 2 - 3i + (-2 + 3i)$$

$$= \underline{\underline{0}} = \underline{\underline{0 + 0i}}$$

① ^{Imp} Division of Compl. Nos.

② ^{Imp} Multiplicative Inverse of Compl. No.

$$5 \times \frac{1}{5} = \{1\} = \frac{1 + 0i}{(1, 0)}$$

Ex

$$Z = 7 - 5i$$

Find Multiplicative Inverse

$$1 = \frac{1}{\cdot} \times \frac{7 + 5i}{\cdot}$$

$$\begin{array}{r} 49 \\ 25 \\ \hline 74 \end{array}$$

$$\frac{1}{7-5i} = \frac{1}{7-5i} \times \frac{7+5i}{7+5i}$$

$$= \frac{7+5i}{7^2 + 5^2} = \frac{7+5i}{74}$$

$$\frac{1}{7-5i} = \frac{7}{74} + \frac{5}{74}i \quad \text{Ans}$$

Short Cut Thus, Multiplicative Inverse of

$$7-5i = \frac{7}{74} + \frac{5}{74}i$$

Let $Z = a+bi$ $Z^{-1} \Rightarrow \frac{1}{Z}$

Multiplicative Inverse of Z

$$= \frac{a}{a^2+b^2} - \frac{b}{a^2+b^2}i$$

Ordered Pair $Z = (a, b)$

Mult. Inverse of Z

Re Im

$$\frac{1}{Z} = \overline{Z} = \left(\frac{a}{a^2+b^2}, \frac{-b}{a^2+b^2} \right)$$

b. $\overline{z_1 z_2} = \bar{z}_1 \cdot \bar{z}_2$

Consider $z_1 = 1 - 2i$ and $z_2 = 4 + 3i$, then:

$$\text{L.H.S} = \overline{z_1 z_2}$$

$$= \overline{(1-2i)(4+3i)}$$

$$= \overline{4+3i-8i-6i^2} \quad (\because i^2 = -1)$$

$$= \overline{10-5i}$$

$$\overline{z_1 z_2} = 10 + 5i \longrightarrow \textcircled{1}$$

$$\text{R.H.S} = \bar{z}_1 \cdot \bar{z}_2$$

$$= (\overline{1-2i})(\overline{4+3i})$$

$$= (1+2i)(4-3i)$$

$$= 4-3i+8i-6i^2$$

$$\bar{z}_1 \cdot \bar{z}_2 = 10 + 5i \longrightarrow \textcircled{2}$$

~~Diff~~

$$\text{L.H.S} = \text{R.H.S}$$

$$\text{Hence, } \overline{z_1 z_2} = \bar{z}_1 \cdot \bar{z}_2$$

Hence, $z_1 z_2 = z_1 \bar{z}_2$
proved

c. $\overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z}_1}{\bar{z}_2}$

Consider $z_1 = -1 + 5i$ and $z_2 = 2 + i$

Sol: $\bar{z}_1 = -1 - 5i$, $\bar{z}_2 = 2 - i$

$$\frac{z_1}{z_2} = \frac{-1 + 5i}{2 + i}$$

$$= \frac{-1 + 5i}{2 + i} \times \frac{2 - i}{2 - i}$$

$$= \frac{-2 + i + 10i - 5i^2}{4 - i^2}$$

$$= \frac{3 + 11i}{5}$$

$$\frac{z_1}{z_2} = \frac{3}{5} + \frac{11}{5}i$$

$$\text{L.H.S} = \overline{\left(\frac{z_1}{z_2}\right)} = \frac{3}{5} + \frac{11}{5}i$$

$$\text{L.H.S.} = \overline{\left(\frac{z_1}{z_2}\right)} = \frac{3}{5} - \frac{11}{5}i \rightarrow \textcircled{1}$$

$$\text{R.H.S.} = \frac{\overline{z_1}}{\overline{z_2}} = \frac{-1-5i}{2-i}$$

$$= \frac{(-1-5i)}{2-i} \times \frac{(2+i)}{2+i}$$

$$= \frac{-2-i-10i-5i^2}{4-i^2}$$

$$= \frac{3-11i}{5}$$

$$\frac{\overline{z_1}}{\overline{z_2}} = \frac{3}{5} - \frac{11}{5}i \rightarrow \textcircled{2}$$

As $\text{L.H.S.} = \text{R.H.S.}$ Hence

Thus, $\overline{\left(\frac{z_1}{z_2}\right)} = \frac{\overline{z_1}}{\overline{z_2}}$ 'proved'

Graphical / Geometrical Representation

Graphical / Geometrical Representation of a Complex No.

$$Z = a + bi$$

$$= (a, b)$$

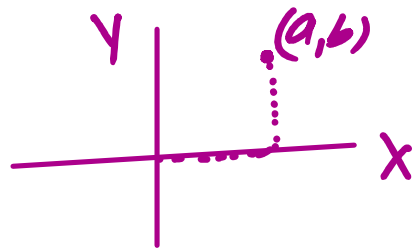
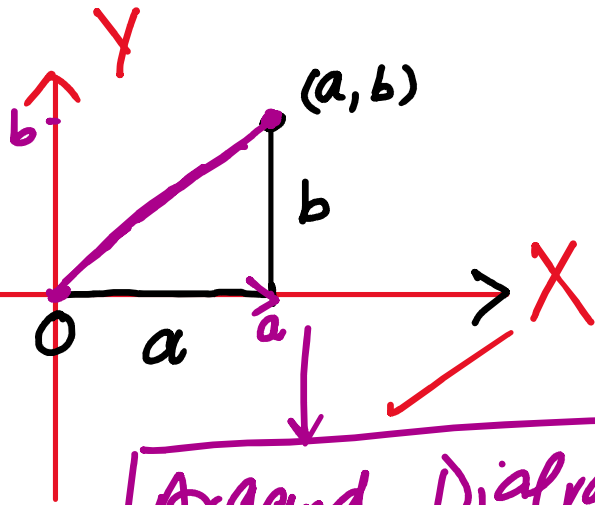
a = Real part
 b = Imaginary part

Z-plane / Complex plane.

2-D Plane

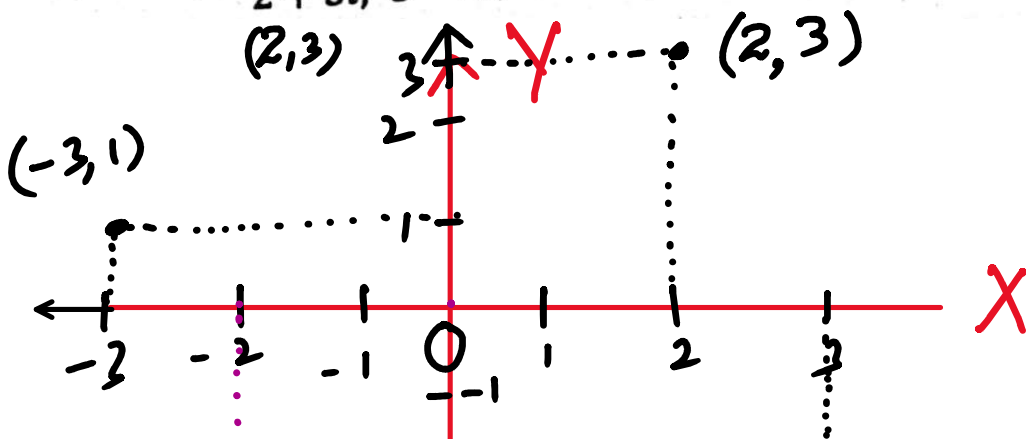
X-axis = Real axis

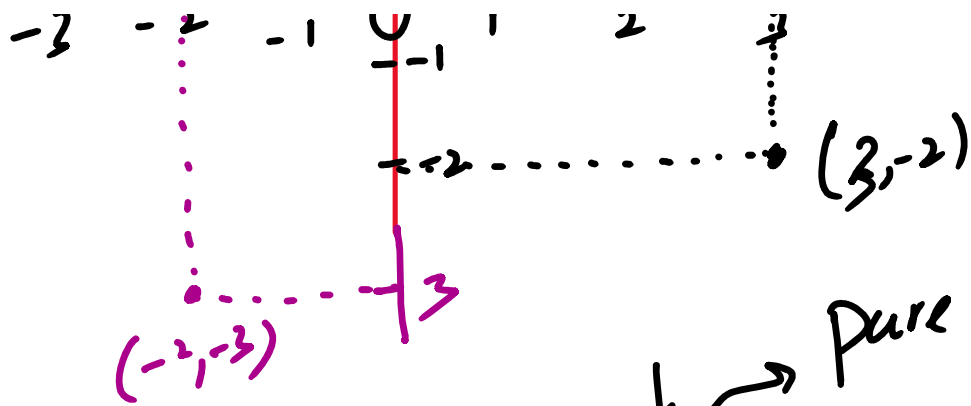
Y-axis = Imaginary axis



Example: Represent the numbers in the complex plane.

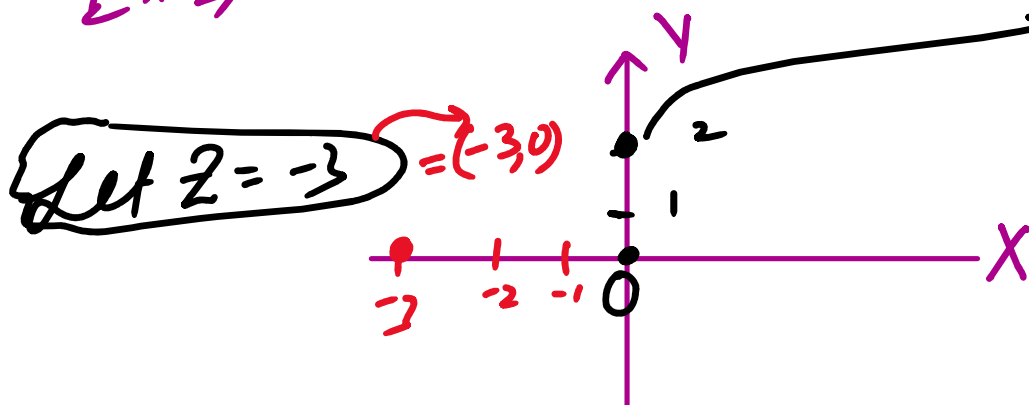
$2 + 3i$, $3 - 2i$, $-3 + i$, $-2 - 3i$





Ex $\Rightarrow$ Let $z = 2i = 0 + 2i = (0, 2)$

Pure Imag. No.



$$z = x + yi$$

$$\operatorname{Re}(z) = x, \quad \operatorname{Im}(z) = y$$

Modulus / Absolute Value $\neq$ $\uparrow$

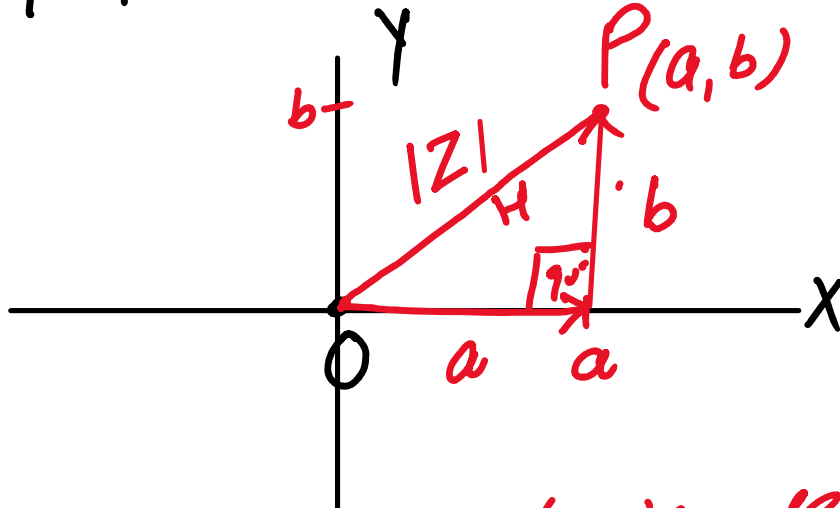
a Complex No.

$$\text{Let } z = a + bi = \underline{\underline{(a, b)}}$$

$$|z| = \sqrt{a^2 + b^2}$$

$$\underline{\underline{z = (a, 0)}}$$

$$|Z| \geq 0$$



$|Z|$ = Length of Hypotenuse
 = Distance of (a, b) from
 origin $(0, 0)$

Example; Find the absolute value of the complex numbers:

a. $5i$

b. 2

c. $4 - 7i$

d. $-3 + 2i$

$$Z = 5i = 0 + 5i$$

$$|Z| = \sqrt{0^2 + 5^2} = \sqrt{5^2} = 5$$

$$Z = -5i = \sqrt{0^2 + (-5)^2} = \sqrt{0^2 + 25} = \sqrt{(-5)^2} = +5$$

d) $Z = -3 + 2i$

$$\sqrt{(-3)^2 + (2)^2} = \sqrt{9 + 4}$$

$$|z| = \sqrt{(-3)^2 + (2)^2} = \sqrt{9+4} = \sqrt{13}$$

If z is a complex number, then following properties holds:

a. $|z| = |-z| = |\bar{z}| = |-\bar{z}|$ b. $\bar{z} \cdot z = |z|^2$

a. Consider a complex number $z = 2 - 5i$, $-z = -2 + 5i$

$$|z| = \sqrt{(2)^2 + (-5)^2} = \sqrt{4+25} = \sqrt{29}$$

$$|-z| = \sqrt{(-2)^2 + (5)^2} = \sqrt{4+25} = \sqrt{29}$$

$$z = 2 - 5i, \bar{z} = 2 + 5i, -\bar{z} = -2 - 5i$$

$$|\bar{z}| = \sqrt{(2)^2 + (5)^2} = \sqrt{4+25} = \sqrt{29}$$

$$|-\bar{z}| = \sqrt{(-2)^2 + (-5)^2} = \sqrt{4+25} = \sqrt{29}$$

Hence, $|z| = |-z| = |\bar{z}| = |-\bar{z}|$

b. $\bar{z} \cdot z = |z|^2$, $z = 2 - 5i, \bar{z} = 2 + 5i$

$$= 17 = |z|^2 = |z^2| \checkmark$$

$$\bar{z} \cdot z = |z|^2 = |z^2| \checkmark$$

$$\begin{aligned} \text{L.H.S} &= \bar{z} \cdot z = (2+5i)(2-5i) \\ &= 4 - (5i)^2 = 4 - 25i^2 \\ &= 4 + 25 = 29 \end{aligned}$$

$$\begin{aligned} \text{R.H.S} &= |z|^2 \\ &= \left(\sqrt{(2)^2 + (-5)^2} \right)^2 \\ &= \left(\sqrt{4 + 25} \right)^2 = \left(\sqrt{29} \right)^2 = 29 \end{aligned}$$

$$\Rightarrow \text{LHS} = \text{RHS} \quad \mu$$

$$\bar{z} \cdot z = |z|^2$$

b. $(1+2i)^{-2}$ Let $z = (1+2i)^{-2}$

$$z = \frac{1}{(1+2i)^2} = \frac{1}{1-4+2(1)(2i)}$$

$$z = (1+2i)^{-2} \quad 1-4+2(1)(2i)$$

$$z = \frac{1}{-3+4i}$$

$$z = \frac{1}{-3+4i} \times \frac{-3-4i}{-3-4i}$$

$$z = \frac{-3-4i}{9+16} = \frac{-3-4i}{25}$$

$$z = -\frac{3}{25} - \frac{4}{25}i$$

$$\text{Re}(z) = \text{Real part of } (1+2i)^{-2}$$

$$= -\frac{3}{25}$$

$$\text{Im}(z) = \text{Imaginary part of } (1+2i)^{-2}$$

$$= -\frac{4}{25}$$

$$\boxed{\text{NOT } i^2}$$



Let $\boxed{(\sqrt{3})^2 \cdot i^2}$
 $3 \cdot (-1) = -3$

$\overline{25}$ $\nearrow$

$$c. \quad Z = \frac{(\sqrt{2} + i\sqrt{3})^2}{1+i} = \frac{(\sqrt{2})^2 + (i\sqrt{3})^2 + 2(\sqrt{2})(\sqrt{3}i)}{1+i}$$

$$Z = \frac{2 - 3 + 2\sqrt{6}i}{1+i}$$

$$Z = \frac{-1 + 2\sqrt{6}i}{1+i}$$

$$Z = \frac{(-1 + 2\sqrt{6}i)}{1+i} \times \frac{(1-i)}{1-i}$$

$$Z = \frac{-1 + 2\sqrt{6}i + i - 2\sqrt{6}i^2}{1+1}$$

$$Z = \frac{(-1 + 2\sqrt{6}) + (2\sqrt{6} + 1)i}{2}$$

$$Z = \frac{-1 + 2\sqrt{6}}{2} + \frac{2\sqrt{6} + 1}{2}i$$

$$Z = \frac{-1 \pm \sqrt{1-2}}{2}$$

Thus, $\left. \begin{aligned} \text{Real part} &= \frac{-1 + \sqrt{2}}{2} \\ \text{Imaginary part} &= \frac{\sqrt{2} + 1}{2} \end{aligned} \right\}$

d. $\frac{(1-i)^2}{(1+i)^2} = \left(\frac{1-i}{1+i} \right)^2$

$$= \left(\frac{(1-i) \times (1+i)}{1+i} \right)^2$$

$$= \left(\frac{1+1}{1+i^2+2i} \right)^2 = \left(\frac{2}{\cancel{1}+2i} \right)^2$$

$$= \left(\frac{\cancel{2}}{2i} \right)^2$$

$$= \left(\frac{1}{i} \right)^2 = \frac{1}{i^2} = \textcircled{-1} + 0i$$

$$= \left(\frac{1}{i}\right) = -i^2$$

$$\left. \begin{array}{l} \text{Real part} = -1 \\ \text{Imag} = 0 \end{array} \right\}$$

Method #02

$$\frac{(1-i)^2}{(1+i)^2} = \frac{1+i^2-2(1)(i)}{1+i^2+2(1)(i)}$$

$$= \frac{1-1-2i}{1-1+2i} = \frac{-2i}{2i} = -1$$

$$\left. \begin{array}{l} \text{Real part} = -1 \\ \text{Imag. part} = 0 \end{array} \right\}$$

$$= z_1 + \underline{\underline{z_2}}$$



Geometrical Representation of Algebraic Operations

Let z_1 is a complex no. represented by point $P(z_1)$

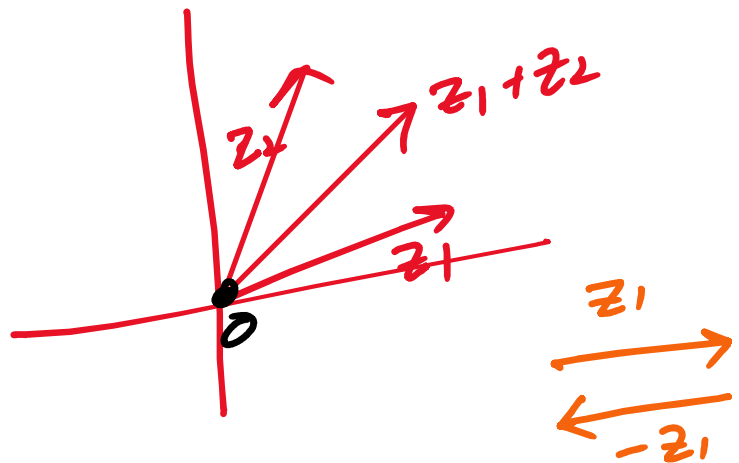
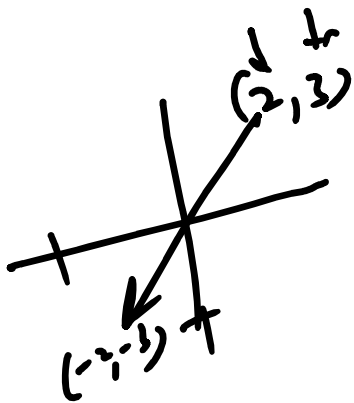
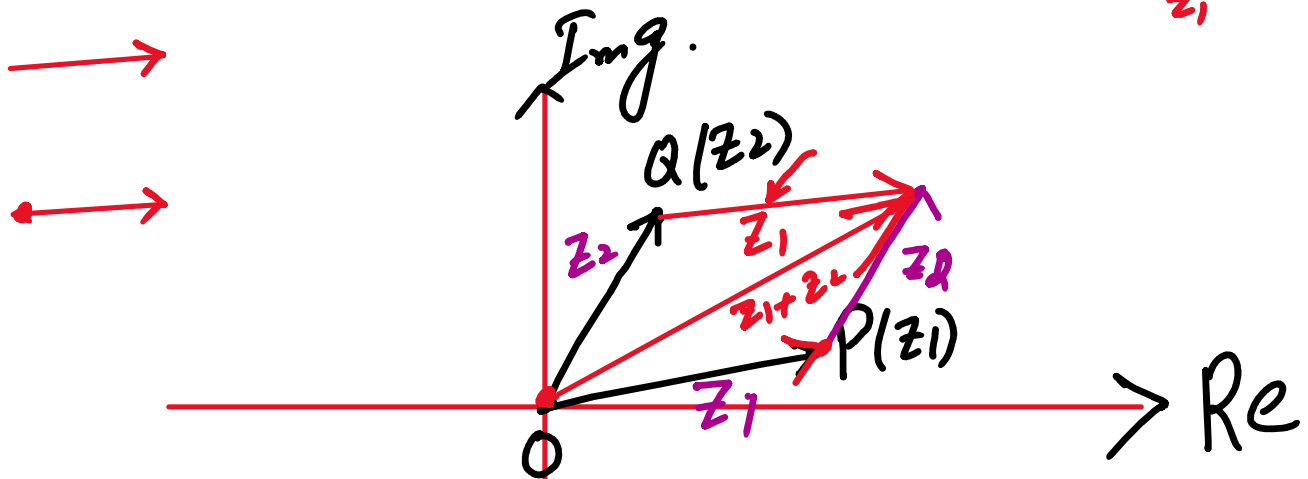
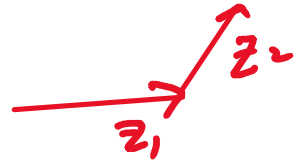
↓

(1) (2)

$z_1 + z_2$

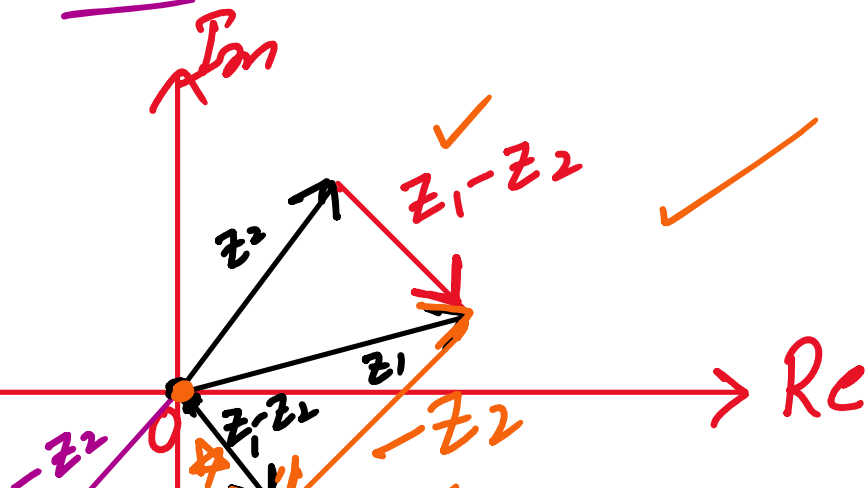
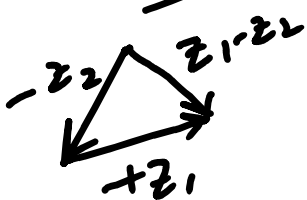
operation

$$Z = Z_1 + Z_2$$



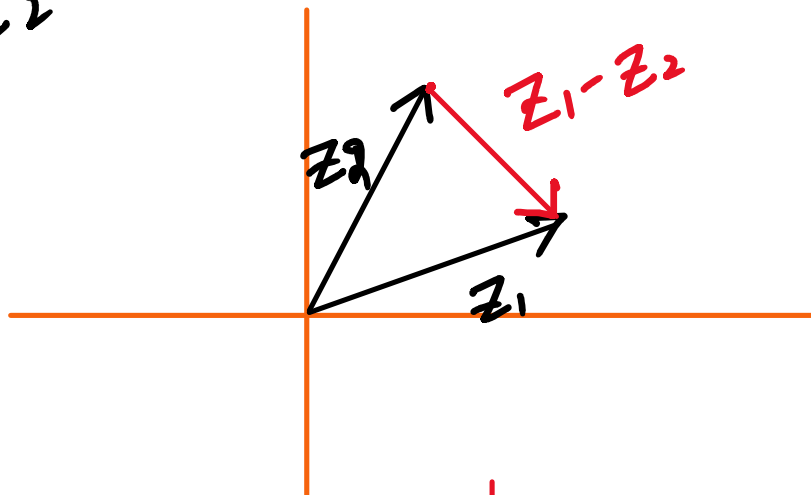
$$= z_1 + (-z_2) = -z_2 + z_1$$

$$\underline{z_1 - z_2}$$

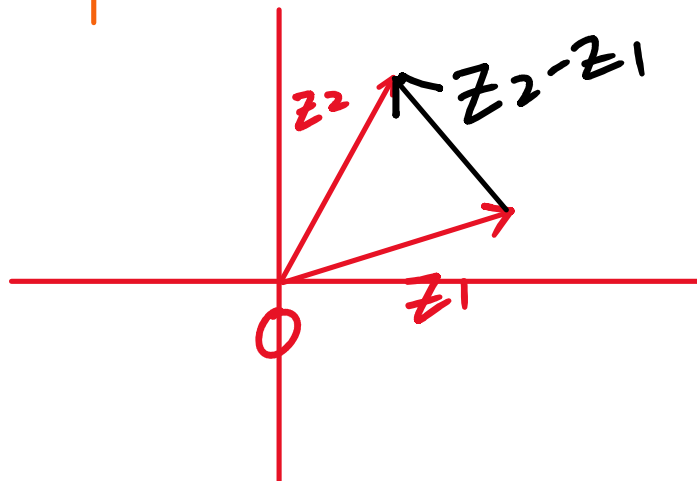




$$z_1 - z_2$$



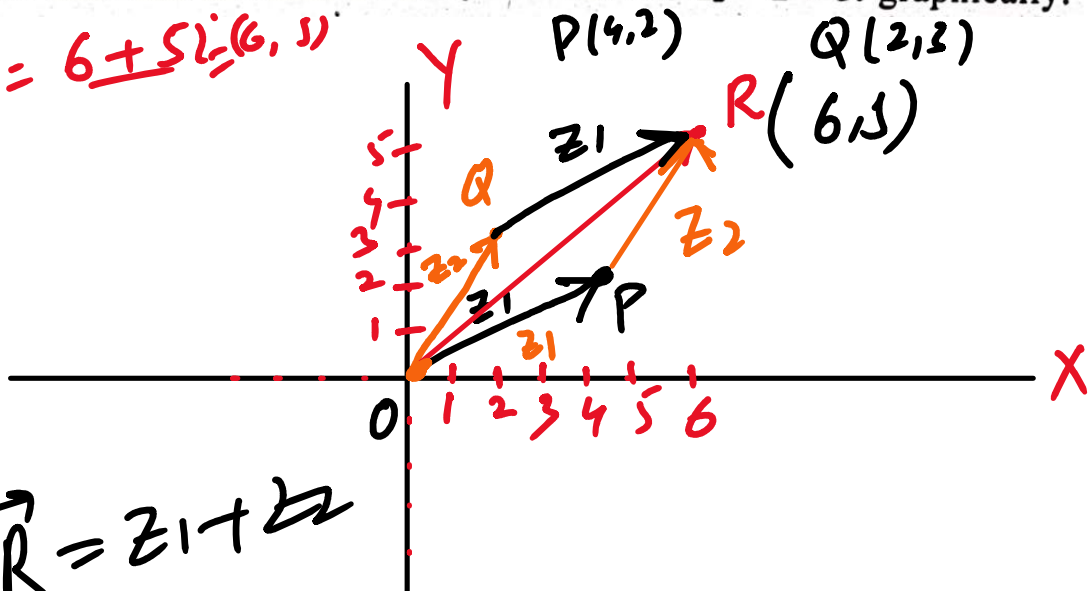
$$z_2 - z_1$$



Example:

Represent sum and difference of $z_1 = 4 + 2i$ and $z_2 = 2 + 3i$ graphically.

$$z_1 + z_2 = 6 + 5i = (6, 5)$$



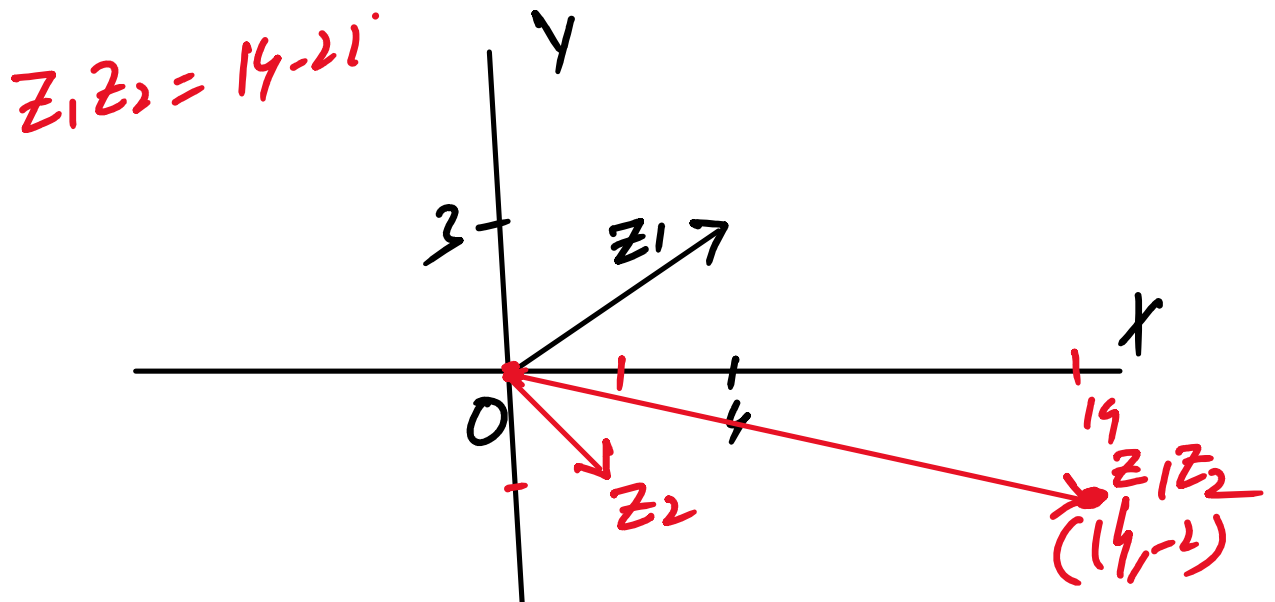
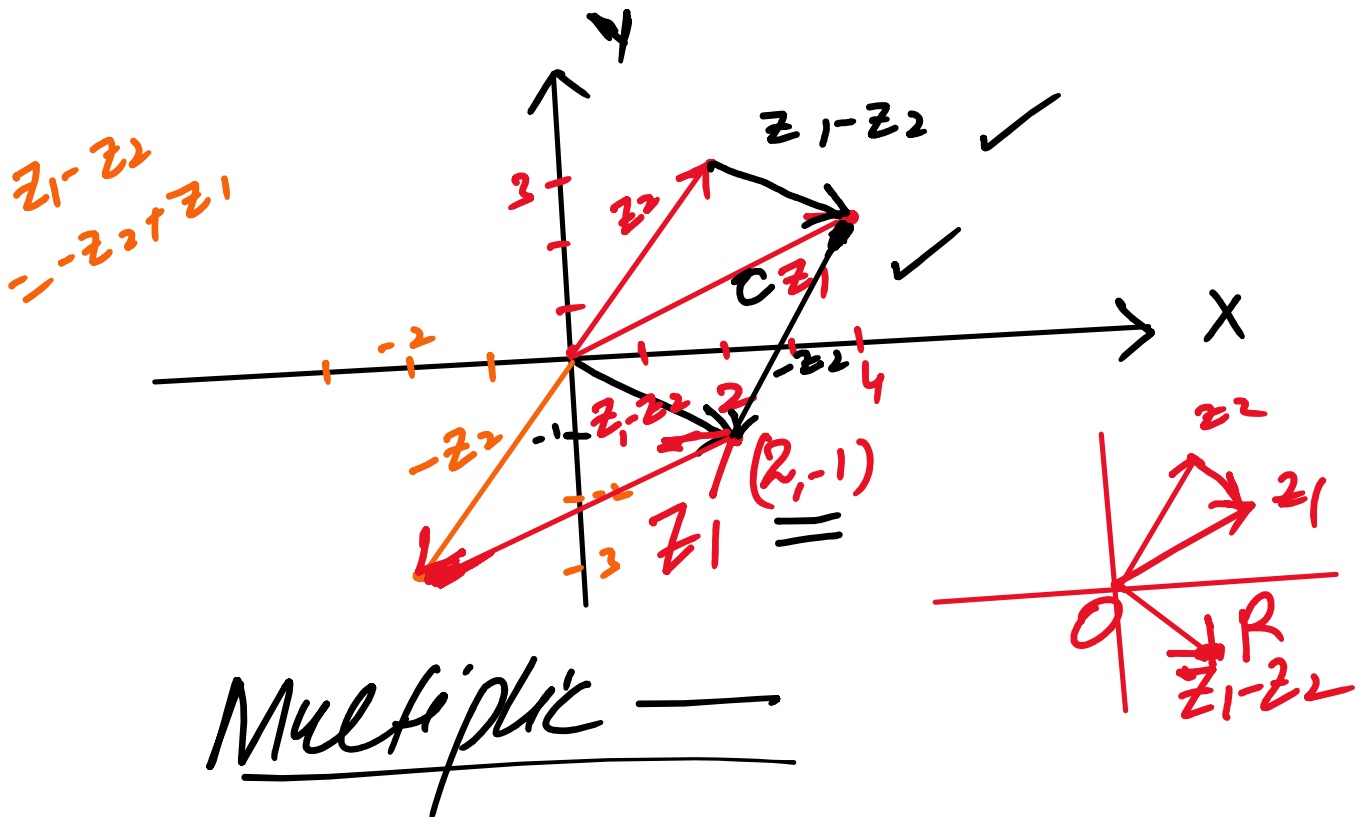
$$\vec{OR} = z_1 + z_2$$

$$\overrightarrow{OR} = z_1 + z_2$$

$$z_1 - z_2 = 4 + 2i - (2 + 3i)$$

$$= 4 + 2i - 2 - 3i$$

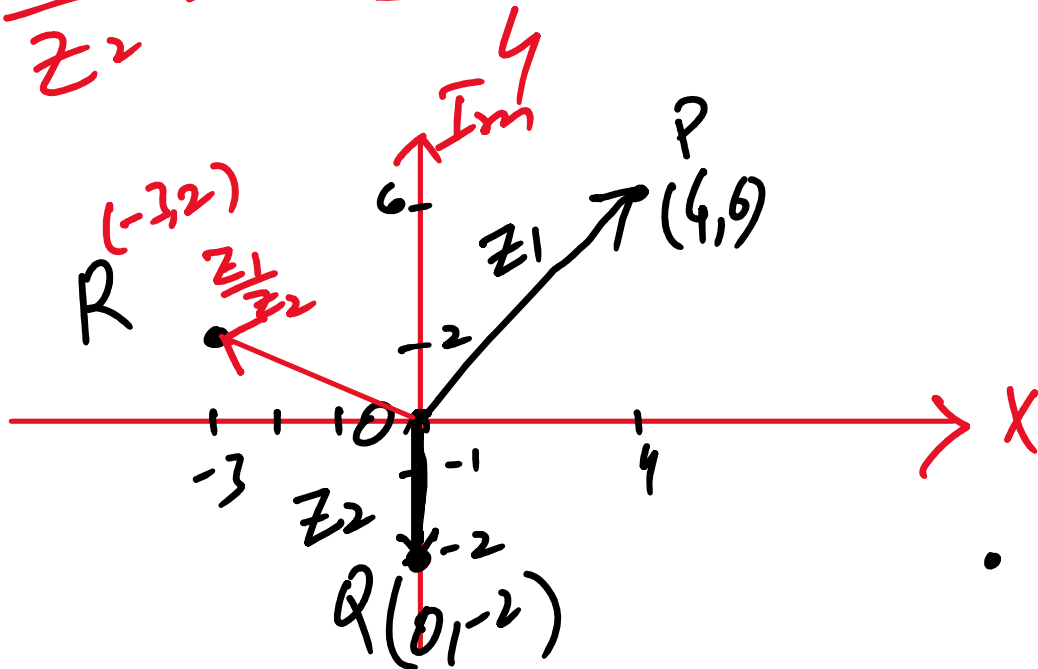
$$z_1 - z_2 = 2 - i$$



Again, consider $z_1 = 4 + 6i$ ✓
and $z_2 = -2i$ then:

$$\frac{z_1}{z_2} = \frac{4+6i}{-2i} \times \frac{2i}{2i} = \frac{8i + 12i^2}{-4i^2}$$

$$\frac{z_1}{z_2} = \frac{-12 + 8i}{4} = \boxed{-3 + 2i}$$



Exercise 1.2

1. Find the additive inverse of each complex number.

a. $-4 + 5i$

b. $-3 - 3i$

c. $5 - 5i$

d. $4i$

2. Show that each pair of complex numbers are multiplicative inverse of each other.

a. $2 + 3i, \frac{2-3i}{13}$ yes

b. $5 - 4i, \frac{5+4i}{41}$ yes

c. $6 + 8i, \frac{3-4i}{50}$ yes

① a) $4 - 5i$ b) $3 + 3i$ c) $-5 + 5i$ d) $-4i$

② c) $(6+8i)\left(\frac{3-4i}{50}\right)$

$$= \frac{18 - 24i + 24i - 32i^2}{50}$$

$$= \frac{18+32}{50} = \frac{50}{50} = \textcircled{1} = 1+0i = \underline{\underline{(1,0)}}$$

3. Find the multiplicative inverse of each complex number.

a. $1+i$

b. $7-3i$ ✓

c. $10-12i$

d. $\frac{2}{5-i} \xrightarrow{z} \frac{1}{z} = \frac{5+i}{2}$

e. $\frac{-i}{2-3i}$

f. $a-bi$

b) Let $z = 7-3i$

To find $z^{-1} = \frac{1}{z}$ = Multiplicative inverse of z

$$\frac{1}{z} = \frac{1}{7-3i} = \frac{1}{7-3i} \times \frac{7+3i}{7+3i}$$

$$\frac{1}{z} = \frac{7+3i}{49+9} = \frac{7}{58} + \frac{3}{58}i$$

... inverse of $7-3i$

z 4771
 Hence, Multiplicative inverse of $7-3i$ is

$$\frac{7}{58} + \frac{3}{58}i \quad \text{Ans}$$

e. $\frac{-i}{2-3i}$

Let $z = \frac{-i}{2-3i}$

$$\frac{1}{z} = \frac{2-3i}{-i}$$

$$= \frac{2-3i}{-i} \times \frac{i}{i}$$

$$= \frac{2i - 3i^2}{-i^2}$$

$$= \frac{2i + 3}{1} = \boxed{3 + 2i}$$

Hence mult. inverse of $\frac{-i}{2-3i}$ is

$$\boxed{3+2i} \quad \text{Ans}$$

d. $\frac{2}{5-i}$

let $z = \frac{2}{5-i}$

$$\frac{1}{z} = \frac{5-i}{2} \quad \text{Ans}$$

f. $a-bi$

let $z = a-bi$

$$\frac{1}{z} = \frac{1}{a-bi}$$

$$= \frac{1}{a-bi} \times \frac{a+ib}{a+ib}$$

$$\boxed{\frac{1}{z} = \frac{a+ib}{a^2+b^2}}$$

$a \quad b \quad i$

$$\frac{1}{z} = \frac{a}{a^2+b^2} + \frac{b}{a^2+b^2}i$$

4. Find the product of each complex number and its conjugate.

a. 4

b. $1-i$

c. $7i$

d. $6-2i$

e. $10+9i$

f. $-4-11i$

4) a) let $z = 4$, $\bar{z} = 4$
 $z\bar{z} = 16$ As

f) let $z = -4-11i$
 $\bar{z} = -4+11i$

$$\begin{aligned} z\bar{z} &= (-4-11i)(-4+11i) \\ &= (-4)^2 - (11i)^2 \\ &= 16 - 121i^2 \\ &= 16 - 121(-1) = 137 \end{aligned}$$

$$= 16 + 121 = 137$$

5. If $z_1 = 1 - 2i$ and $z_2 = 2 + i$:

a) $\rightarrow$ Do yourself as it
is a ditto repetition of Example 4

b. Find:

i. $|z_1 + z_2|$

ii. $|z_1 z_2|$

iii. $\left| \frac{z_1}{z_2} \right|$

5. If $z_1 = 1 - 2i$ and $z_2 = 2 + i$:

i) $z_1 + z_2 = 3 - i$

$$|Z_1 + Z_2| = \sqrt{(3)^2 + (-1)^2} = \sqrt{9+1} = \sqrt{10} \quad \text{R}$$

iii) $\frac{Z_1}{Z_2} = \frac{1-2i}{2+i}$

$$-1-2i \vee 2-i$$

$$= \frac{1-2i}{2+i} \times \frac{2-i}{2-i}$$

$$= \frac{2-i-4i+2i^2}{4+1}$$

$$= \frac{2-5i-2}{5}$$

$$\frac{z_1}{z_2} = -i = 0-i$$

$$\left| \frac{z_1}{z_2} \right| = \sqrt{(0)^2 + (-1)^2} = 1$$

6. Represent the numbers in the complex plane.

a. $-1-3i$

b. $2+4i$

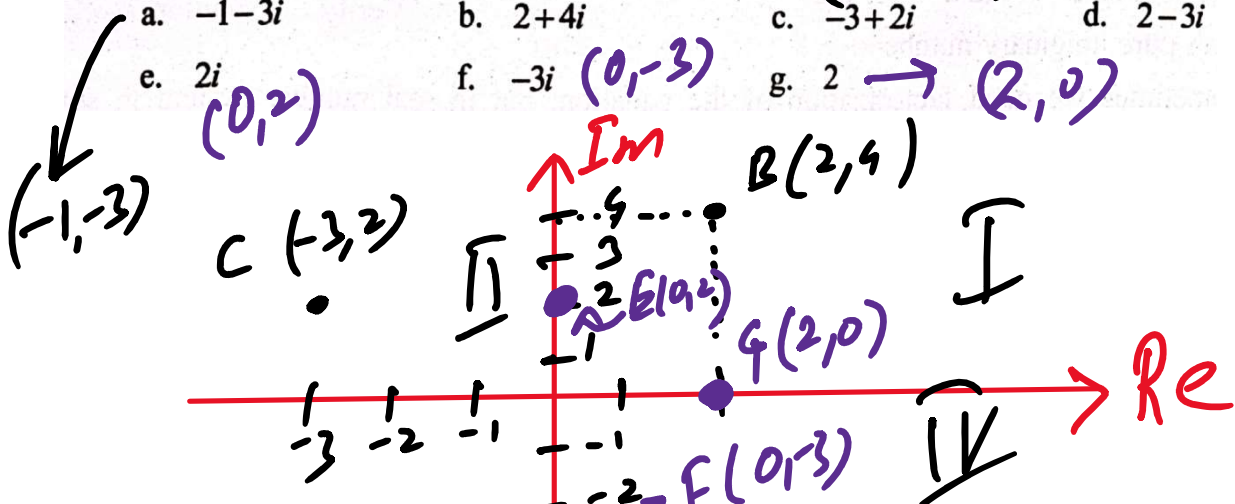
c. $-3+2i$

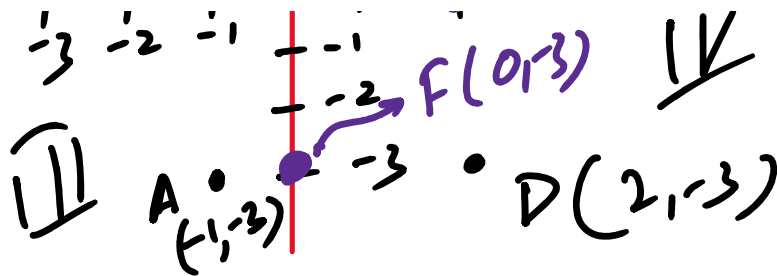
d. $2-3i$

e. $2i$

f. $-3i$

g. $2 \rightarrow (2,0)$





7. Separate into real and imaginary parts of each complex number.

a. $(\sqrt{2} - \sqrt{3}i)^2$

b. $(\sqrt{2} + i)^2$

c. $\frac{(2+3i)^2}{1-3i}$

d. $\left[\frac{1}{2} + \frac{\sqrt{3}}{2}i \right]^2$

e. $\frac{1-i}{(i)^2}$

f. $\frac{1}{i(1-i)^2}$

g. $\frac{(1+i)^2}{(1-2i)^2}$

$$\begin{aligned} d) &= \left(\frac{1 + \sqrt{3}i}{2} \right)^2 \\ &= \frac{(1 + \sqrt{3}i)^2}{(2)^2} \\ &= \frac{1 + (\sqrt{3}i)^2 + 2(1)(\sqrt{3}i)}{4} \\ &= \frac{1 - 3 + 2\sqrt{3}i}{4} \\ &= \frac{-2 + 2\sqrt{3}i}{4} \end{aligned}$$

$$= -\frac{2}{4} + \frac{4\sqrt{3}i}{4(2)}$$

$$= -\frac{1}{2} + \frac{\sqrt{3}i}{2}$$

Hence,

$$\boxed{\begin{array}{l} \text{Real part} = -\frac{1}{2} \\ \text{Imaginary part} = \frac{\sqrt{3}}{2} \end{array}} \quad \text{A}$$

f. $\frac{1}{i(1-i)^2}$

g. $\frac{(1+i)^2}{(1-2i)^2}$

f) Let $Z = \frac{1}{i(1-i)^2}$

$$= \frac{1}{i(1+i^2-2(1)(i))}$$

$$= \frac{1}{i(1-1-2i)} = \frac{1}{-2i^2}$$

$$= \frac{i(1-1-2i)}{-2(-1)} = \frac{1}{2} + 0i$$

$$\boxed{\begin{array}{l} \text{Real part} = \frac{1}{2} \\ \text{Imaginary part} = 0 \end{array}}$$

Let

$$Z = \frac{(1+i)^2}{(1-2i)^2} = \left(\frac{1+i}{1-2i} \right)^2$$

$$= \left(\frac{1+i}{1-2i} \times \frac{1+2i}{1+2i} \right)^2$$

$$= \left(\frac{1+2i+i+2i^2}{1+4} \right)^2$$

$$= \left(\frac{1-2+3i}{5} \right)^2$$

$$\begin{aligned}
 &= \left(\frac{-1 + 3i}{5} \right)^2 \\
 &= \frac{(-1)^2 + (3i)^2 - 2(1)(3i)}{25} \\
 &= \frac{1 - 9 - 6i}{25} \\
 &= \frac{-8 - 6i}{25} \\
 &= -\frac{8}{25} - \frac{6}{25}i
 \end{aligned}$$

Thus Real part of $z = -\frac{8}{25}$.

Imag. $\dots = -\frac{6}{25}i$ b

2nd Method

long.

2nd method

$$Z = \frac{(1+i)^2}{(1-2i)^2} \Rightarrow Z = \frac{\cancel{1} + 2i}{1-4-2(2i)}$$

$$Z = \frac{2i}{-3-4i}$$

$$Z = \frac{2i}{-3-4i} \times \frac{-3+4i}{-3+4i}$$

$$Z = \frac{-6i-8}{9+16} = \frac{-8-6i}{25}$$

$$Z = -\frac{8}{25} - \frac{6}{25}i$$

$$\begin{aligned} \text{Re}(Z) &= -8/25 \\ \text{Im}(Z) &= -6/25 \end{aligned}$$

~~$$\arg(z) = -\theta/\pi$$~~

8. Taking any complex number and show that:

- a. $z \cdot \bar{z}$ is a real number. b. $z^2 + (\bar{z})^2$ is a real number.
 c. $(z - \bar{z})^2$ is a real number. d. $|z|$ and $|\bar{z}|$ are real numbers.
 e. $z^2 - (\bar{z})^2$ is an imaginary number.

a) let $z = a + bi$, $a, b \in \mathbb{R}$

$$\bar{z} = a - bi$$

$$\begin{aligned} z\bar{z} &= (a + bi)(a - bi) \\ &= a^2 - (bi)^2 \\ &= a^2 - b^2 i^2 \end{aligned}$$

$$\boxed{z\bar{z} = a^2 + b^2}$$

As $a^2 + b^2 \in \mathbb{R}$

Hence, $z\bar{z}$ is a Real Number.

b. $z^2 + (\bar{z})^2$ is a real number.

2 , , 2

$$\begin{aligned}
 z^2 + (\bar{z})^2 &= (a+bi)^2 + (a-bi)^2 \\
 &= a^2 + (bi)^2 + 2(a)(bi) + a^2 + (bi)^2 - 2(a)(bi) \\
 &= a^2 - b^2 + \cancel{2abi} + a^2 - b^2 - \cancel{2abi} \\
 &= 2a^2 - 2b^2
 \end{aligned}$$

$$z^2 - (\bar{z})^2 = 2(a^2 - b^2)$$

$$\text{As } 2(a^2 - b^2) \in \mathbb{R}$$

Hence $z^2 - (\bar{z})^2$ is a Real no.

c. $(z - \bar{z})^2$ is a real number.

$$\begin{aligned}
 (z - \bar{z})^2 &= (a+bi - (a-bi))^2 \\
 &= (a+bi - a+bi)^2
 \end{aligned}$$

$$= (a+bi - a+bi)^2$$

$$= (2bi)^2$$

$$= 4b^2i^2$$

$$(z - \bar{z})^2 = -4b^2 \quad (\because i^2 = -1)$$

As $-4b^2 \in \mathbb{R}$

Hence $(z - \bar{z})^2$ is a Real no.

d. $|z|$ and $|\bar{z}|$ are real numbers.

Let $z = a+bi$, $\bar{z} = a-bi$

$$|z| = \sqrt{a^2 + b^2}, \quad |\bar{z}| = \sqrt{a^2 + (-b)^2}$$

$$|\bar{z}| = \sqrt{a^2 + b^2}$$

$$= \sqrt{a^2 + b^2}$$

$$\Rightarrow |Z| = |\bar{Z}| = \sqrt{a^2 + b^2}$$

$$\text{As } \sqrt{a^2 + b^2} \in \mathbb{R}$$

Hence, $|Z|$ & $|\bar{Z}|$ are real nos.

e. $z^2 - (\bar{z})^2$ is an imaginary number.

$$\text{Let } z = a + bi, \quad \bar{z} = a - bi$$

$$z^2 - (\bar{z})^2 = (a + bi)^2 - (a - bi)^2$$

$$= a^2 - b^2 + 2abi - (a^2 - b^2 - 2abi)$$

$$= \cancel{a^2} - \cancel{b^2} + 2abi - \cancel{a^2} + \cancel{b^2} + 2abi$$

$$z^2 - (\bar{z})^2 = 4abi \rightarrow \text{Pure imaginary number}$$

Hence, $z^2 - (\bar{z})^2$ is an imaginary number

9. Represent sum and difference of complex numbers graphically.

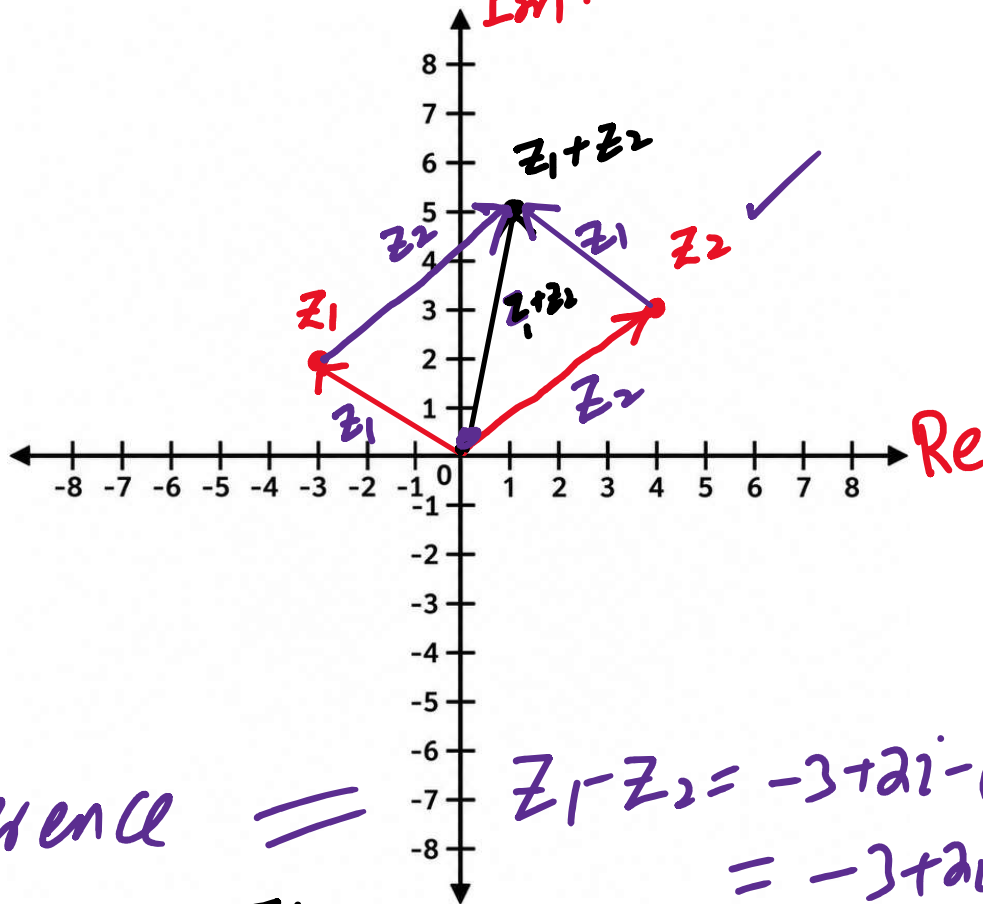
i) (i) $z_1 = 5 + 3i$ and $z_2 = 2 - 3i$

(ii) $z_1 = -3 + 2i$ and $z_2 = 4 + 3i$

Sum = $z_1 + z_2 = 1 + 5i = (1, 5)$
Im.

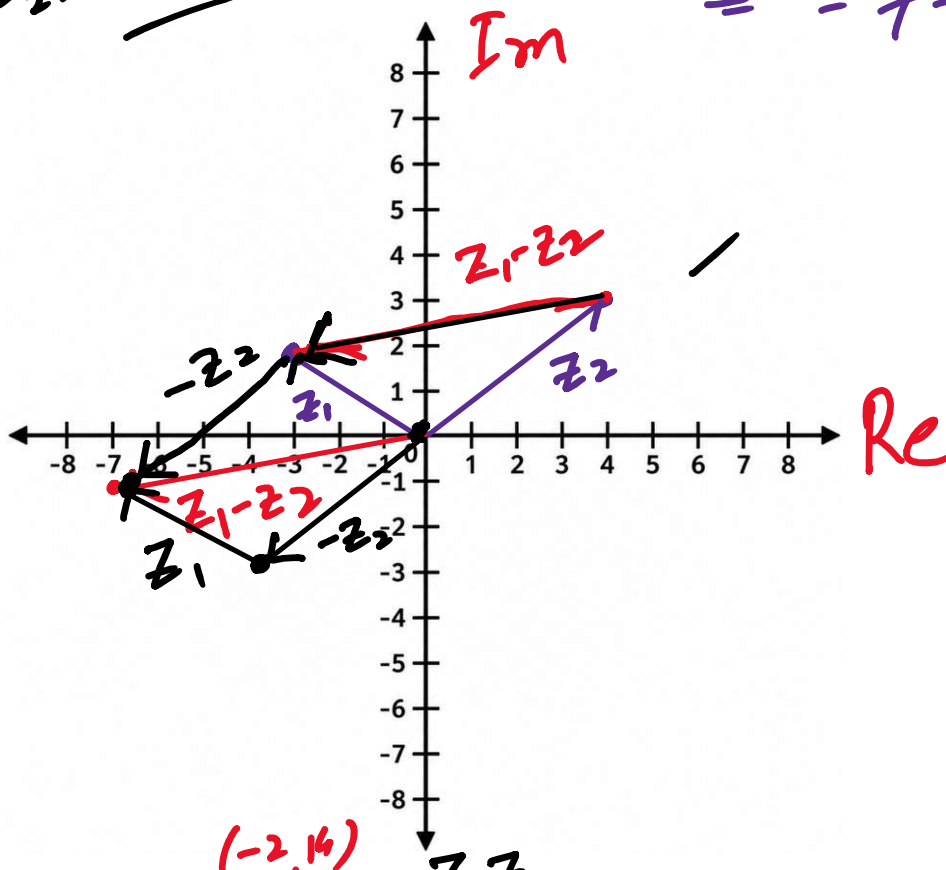
$(-3, 2)$

$(4, 3)$



Difference = $z_1 - z_2 = -3 + 2i - (4 + 3i)$
 $= -3 + 2i - 4 - 3i$
 $= -7 - i = (-7, -1)$

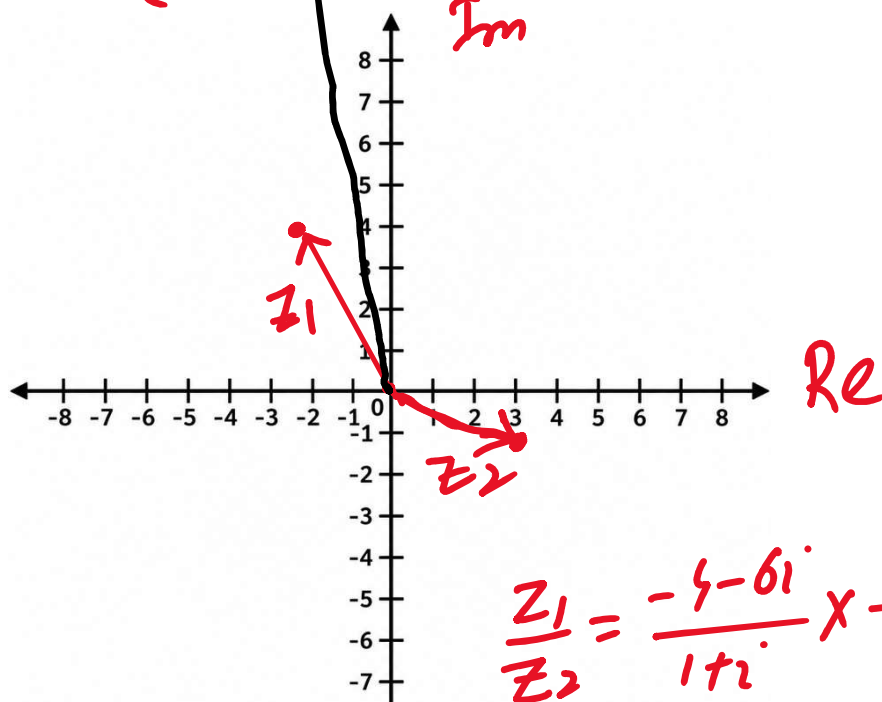
$z_1 - z_2 = -z_2 + z_1$



10. Represent product of complex numbers graphically.

- (i) $z_1 = 4 + 2i$ and $z_2 = -2 + 3i$ (ii) $z_1 = -2 + 4i$ and $z_2 = 3 - i$

$$z_1 z_2 = (-2 + 4i)(3 - i) = -6 + 2i + 12i + 4 = -2 + 14i \quad (-2, 14)$$



$$\frac{z_1}{z_2} = \frac{-4 - 6i}{1 + i} \times \frac{1 - i}{1 - i} = \frac{-10 - 2i}{2} = -5 - i = (-5, -1)$$

11. Represent $z_1 \div z_2$ graphically when:

- (i) $z_1 = 6 - 4i$ and $z_2 = 3$ (ii) $z_1 = -4 - 6i$ and $z_2 = 1 + i$

